Aaron Lopez Mendoza

Full-Stack Software Engineer | Databases • Local AI • Linux
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Full-stack software engineer with 3+ years building web applications and database projects, spanning full-stack development with Python/Django and PostgreSQL, local AI model deployment, and agile teamwork with a focus on secure coding practices.

PROJECTS

Reservation Web Platform | Python, Django, PostgreSQL, Leaflet, HTML/CSS — **Mexico City, Mexico**
Festival de las Luciérnagas, UNAM — Team Project, Scrum Master — *2026*

- Built a full-stack platform in Python (Django), HTML and CSS over a 10-table PostgreSQL schema, handling dynamic cabin and campsite inventory sourced from government datasets, with an interactive Leaflet map and validation of availability, dates and business rules.
- Served as Scrum Master for a 5-person team, coordinating weekly sprints and removing blockers under Scrum, Kanban and Extreme Programming.
- Applied OWASP Secure Coding practices (password hashing, role-based access control), added email notifications for booking confirmation and password recovery, and structured the system with design patterns (MTV, Singleton, Strategy, Observer) for a modular, maintainable codebase.

Local AI Deployment & Administration | Ollama, Linux, Windows, macOS — **Mexico City, Mexico**
Personal Project — *2026*

- Deployed and administered the 10 most widely used open models (Llama, Mistral, Gemma) on Windows, Linux and macOS, handling dependencies and runtime environments.
- Benchmarked all 10 models on 4,000-token workloads, comparing latency, resource usage and response quality to select the best fit per hardware profile.
- Automated project scaffolding through the local API, cutting setup time from ~1 hour to 5 minutes (12x faster) with zero cloud dependency; built custom skills and workflows in a self-hosted home lab.

Event Management System | PostgreSQL, SQL, Triggers — **Mexico City, Mexico**
Database Fundamentals, UNAM — *2025*

- Designed and implemented a 32-table PostgreSQL model with primary and foreign keys to enforce referential integrity across the schema.
- Wrote 25+ multi-table SQL queries (JOINs, subqueries, constraints) and triggers to automate database behavior and consolidate cross-table reporting.
- Built a visual reporting layer to surface query results, and fixed referential integrity errors to improve data consistency.

Shamir's Secret Sharing Scheme | Python, cryptography — **Mexico City, Mexico**
Modeling and Programming, UNAM — *2025*

- Implemented Shamir's Secret Sharing in Python to protect text files, splitting a key into shares recoverable only above a minimum threshold.

Additional Academic Projects | C, C++, MPI, Python, Docker — **Mexico City, Mexico**
Distributed Computing and Programming Courses, UNAM — *2025*

- Parallelized workloads in C/C++ with MPI, achieving a 6.8x speedup on 8 processes (85% parallel efficiency) through efficient memory handling and complexity analysis.
- Python application consuming external weather APIs; Docker containers for deploying PostgreSQL and reproducible development environments.

EDUCATION

Universidad Nacional Autónoma de México (UNAM) — **Mexico City, Mexico**

- **Major:** B.Sc. in Computer Science — *Graduation date: June 2027*
- **Cumulative GPA:** 9.25/10 (~3.70/4.0, US scale)
- **Relevant Coursework:** Data Structures, OOP, Functional Programming, Databases, Distributed Computing, Software Engineering, Computer Security.

TECHNICAL SKILLS

Programming: Python, Java, C, C++, Kotlin, Racket, Assembly, JavaScript, HTML, CSS
Databases: PostgreSQL, SQL, relational design, referential integrity
Technologies: Git, Docker, Linux, Ollama, Django, Leaflet, MPI
Concepts: Data structures, OOP, functional programming, distributed computing, networking and security fundamentals (Cisco, VPNs, OWASP)
Languages: Spanish (native) | English (Intermediate — technical documentation)